

Semester: July 2025 – November 2025

Duration: 3 Hrs.

Maximum Marks: 100

Examination: ESE Examination

Class: LY

Semester: VII (SVU 2020)

Programme code: 01

 Programme: **B.Tech COMP**

 Institute/School/Department: K. J. Somaiya School
 of Engineering

 Name of the department:
 COMP

Course Code: 116U01C701

Name of the Course: Software Architecture and Design Thinking

 Instructions: 1) Draw neat diagrams 2) All questions are compulsory
 3) Assume suitable data wherever necessary

Que. No.	Question	Max. Marks
		20
Q1	Solve any FOUR	5
i)	How does "Lack of shredding" affect software architecture?	5
ii)	Compare Ambiguity, Precision and Accuracy with respect to Software Architecture giving suitable example	5
iii)	Explain need of connectors in software architecture	5
iv)	What do you understand by Descriptive Software Architecture and Perspective Software architecture?	5
v)	What are the basic architectural elements of A Software Model? With an example explain the same	5
vi)	Differentiate between Goals of Analysis and Scope of Analysis	5

Que. No.	Question	Max. Marks
		10
Q2 A	Solve the following	5
i)	Explain the Role of a Software Architect in development of a software system	5
ii)	What are reasons for Architectural Degradation?	
	OR	
Q2 A	Explain in detail ANY FIVE structural design issues giving example of each.	10
Q2 B	Solve any One	10
i)	With reference to Architectural Analysis Goals, explain 4 Cs	
ii)	Explain 5 Phase Process of Software Architecture Design	
Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	You are appointed as Software Architect for designing a banking website . Explain with suitable diagram appropriate Software Architecture. Justify the architectural style used in designing.	10
ii)	Explain commonly used view points of Banking website	10
iii)	How are NFPs taken care into software architecture of Linux operating system	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Justify the need of Identifying Drivers in the Research Stage of Design Thinking Explain with examples of each type of the drivers	03 07
ii)	Compare "Thinking in Signs" with "Thinking in Images" with reference to the Refinement stage of Design Thinking. Give Examples of each	06 04
iii)	With reference to the Refinement stage of Design Thinking, explain giving examples, key forms of Appropriation	10

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Justify the need of "Feedback" stage in the Design Thinking Process	5
ii)	With suitable examples, explain Basic Design Directions	5
iii)	What do you understand by Python Philosophy with reference to Design Thinking Process?	5
iv)	Explain in brief the various types of prototypes used in Design Thinking Process	5
v)	With merits and limitations of each, explain the different sampling methods used in Research stage of Design Thinking Process?	5
vi)	Explain in brief the methods of identifying drivers in research stage of the Design Thinking process.	5